

Assignment 2.

1. SELECT DISTINCT (department)
FROM students;

TABLE students

department
IT
HR
Finance

2. SELECT department ,
AVG (age) AS average_age
FROM students
GROUP BY department;

TABLE students

department	avg - age
IT	20.5
HR	22
Finance	23.

3. SELECT department ,
COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

department	student_count
IT	2
HR	2

4. SELECT *
FROM students
WHERE age BETWEEN 21 AND 23;

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT *
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department,
SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

department	total_credits
IT	11

7. ~~SELECT~~ course_id, course_name

7. SELECT course_id ,
 course_name ,
 department ,
 credits
 FROM courses
 WHERE credits <> 4 ;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id , ~~ex~~
 course_name ,
 credits
 FROM courses
 ORDER BY credits DESC
 LIMIT 3 ;

course_id	course_name	credits
102	Python	4
103	Data science	4
101	SQL Basics	3

9. SELECT MAX (grade) AS max_grade ,
 MIN (grade) AS min_grade ,
 Avg (grade) AS avg_grade
 FROM enrollments ;

max_grade	min_grade	avg_grade
90	78	84.6

10. SELECT course_id,
COUNT(*) AS enrollment_count
FROM enrollments
Group By course_id;

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
Group By department;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department,
AVG(salary) AS avg_salary
FROM Salaries
Group By departments
HAVING AVG(salary) > 55000

department	avg - salary
IT	61 000
Finance	70 000

13. SELECT employee - id,
name,
salary,
bonus,
salary + bonus AS total - compensation
FROM salaries
WHERE salary + bonus > 60 000;

employee - id	name	salary	bonus	total - compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14 SELECT department
SUM (budget) AS total - budget,
AVG (budget) AS avg - budget
FROM Projects
GROUP BY department
HAVING AVG (budget) > 70 000;

department	total - budget	avg - budget
IT	270 000	135 000
Finance	80 000	80 000


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15 SELECT project_id,
           project_name,
           department,
           budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000 AND
      department <> 'Marketing';

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Project - id	project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	60000
5	HR Portal	HR	50000